MSG No. 194.271 $P6_3/m'm'c'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3'm'.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3,5,7,8,9,13,15,17,19,20,21]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,4,6,10,11,12,14,16,18,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,3,5,10,11,12,14,16,18,19,20,21]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,4,6,7,8,9,13,15,17,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,3,5,10,11,12,14,16,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[2,4,6,7,8,9,13,15,17,22,23,24]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[1,3,5,10,11,12,14,16,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[2,4,6,7,8,9,13,15,17,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m'.`

No.	position	mapping
1	$[0, 0, z]$	`[1,3,5,19,20,21]`
2	$\left[0, 0, z + \frac{1}{2}\right]$	`[2,4,6,22,23,24]`
3	$[0, 0, -z]$	`[7,8,9,13,15,17]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[10,11,12,14,16,18]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m'.`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,3,5,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[2,4,6,22,23,24]`
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,8,9,13,15,17]`
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[10,11,12,14,16,18]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2/m'.`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,7,13,19]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,10,14,22]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,8,15,20]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,11,16,23]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[5,9,17,21]`
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,12,18,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,11,16,19]`
2	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[2,9,17,22]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,12,18,20]`
4	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[4,7,13,23]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,10,14,21]`
6	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[6,8,15,24]`

Table 9: Wyckoff site: `12i`, site symmetry: `.2.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,10]`
3	$[0,\ x,\ 0]$	`[3,8]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11]`
5	$[-x,\ -x,\ 0]$	`[5,9]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,12]`
7	$[-x,\ 0,\ 0]$	`[13,19]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[14,22]`
9	$[0,\ -x,\ 0]$	`[15,20]`
10	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[16,23]`
11	$[x,\ x,\ 0]$	`[17,21]`
12	$\left[0,\ x,\ \frac{1}{2}\right]$	`[18,24]`

Table 10: Wyckoff site: `12j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,16]`
2	$\left[x-y,\ x,\ \frac{3}{4}\right]$	`[2,17]`
3	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[3,18]`
4	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[4,13]`
5	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	`[6,15]`
7	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	`[7,23]`

continued ...

Table 10

No.	position	mapping
8	$\left[y,\ x,\ \frac{3}{4}\right]$	[8,24]
9	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[9,22]
10	$\left[x,\ x-y,\ \frac{1}{4}\right]$	[10,21]
11	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	[11,19]
12	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[12,20]

Table 11: Wyckoff site: 12k, site symmetry: .m'.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,19]
2	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[2,22]
3	$[-2x,\ -x,\ z]$	[3,20]
4	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[4,23]
5	$[x,\ -x,\ z]$	[5,21]
6	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[6,24]
7	$[-x,\ -2x,\ -z]$	[7,13]
8	$[2x,\ x,\ -z]$	[8,15]
9	$[-x,\ x,\ -z]$	[9,17]
10	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[10,14]
11	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[11,16]
12	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[12,18]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$[-y,\ x-y,\ z]$	[3]
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[4]
5	$[-x+y,\ -x,\ z]$	[5]
6	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	[6]
7	$[x-y,\ -y,\ -z]$	[7]
8	$[y,\ x,\ -z]$	[8]
9	$[-x,\ -x+y,\ -z]$	[9]
10	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[10]
11	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[14]
15	$[y,\ -x+y,\ -z]$	[15]
16	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[16]
17	$[x-y,\ x,\ -z]$	[17]
18	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[18]
19	$[-x+y,\ y,\ z]$	[19]

continued ...

Table 12

No.	position	mapping
20	$[-y,\, -x,\, z]$	[20]
21	$[x,\, x-y,\, z]$	[21]
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[24]